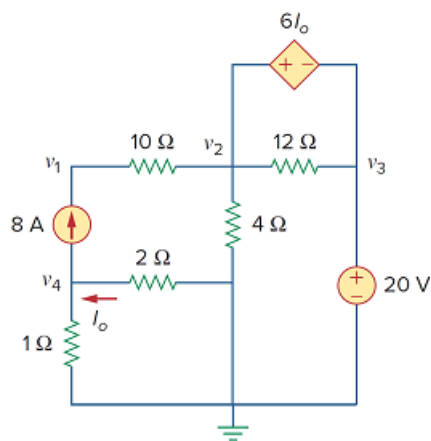


Problem

Find the nodal voltages v_1 through v_4 in the circuit of Fig. 3.120 using *PSpice* or *MultiSim*.

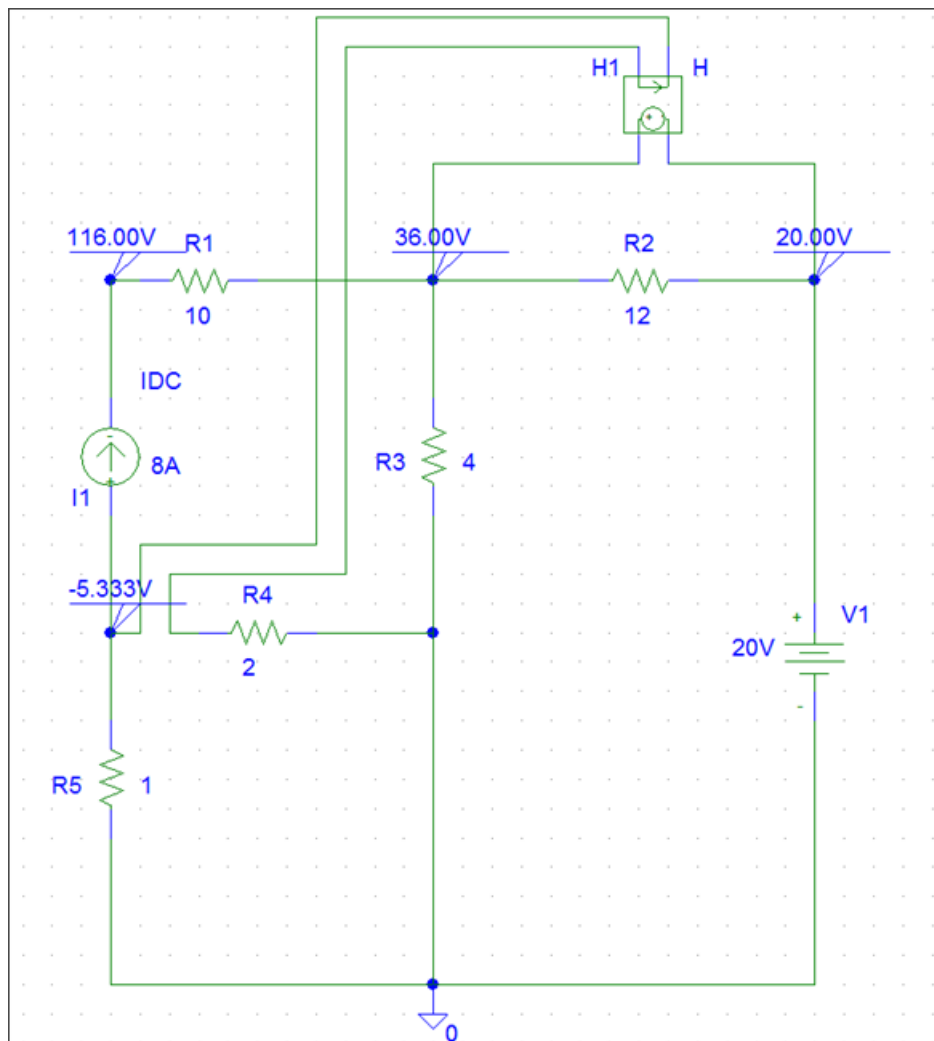
Figure 3.120:



Step-by-step solution

Step 1 of 3

Using PSpice schematics we draw the circuit as shown in figure and simulate it.



Step 2 of 3

* Schematics Netlist *

I_I1 \$N_0001 \$N_0002 DC 8A

V_V1 \$N_0003 0 20V

R_R4 \$N_0004 0 2

R_R3 0 \$N_0005 4

R_R2 \$N_0005 \$N_0003 12

R_R1 \$N_0002 \$N_0005 10

H_H1 \$N_0005 \$N_0003 VH_H1 6

VH_H1 \$N_0004 \$N_0001 0V

R_R5 0 \$N_0001 1

Step 3 of 3

Clearly,

$$V_1 = 116 \text{ V}$$

$$V_2 = 36 \text{ V}$$

$$V_3 = 20 \text{ V}$$

$$V_4 = -5.33 \text{ V}$$